

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2024

Programme: BE

Full Marks: 100

Course: Programming in C (New)

Pass Marks: 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why is C called structured programming language? Which programming language do you prefer to develop a system software and application software? Explain with reasons. 7
b) Define an algorithm. Draw a flowchart to find the factorial of a given number. 8
2. a) What are formatted input/output functions? Explain with examples. 7
b) Define datatypes. Explain basic data types with examples. 8
3. a) Explain any two conditional statements in detail with syntax and flowchart. 8

OR

Write a program to find the n^{th} term of a Fibonacci series 0, 1, 1, 2, 3, 5,

- b) Write a program that prompts the user to enter n numbers from the keyboard, find the sum of even numbers and display the sum. 7
4. a) What are the advantages of using array in programming? Explain different ways to initialize one dimensional array with examples. 8

OR

Write a program to find the product of two $m \times n$ and $n \times m$ matrices and display the resulted matrix.

- b) Explain Formal arguments and Actual arguments with an example program. 7
5. a) Using user defined function write a program to input n numbers in an one dimensional array and find the average of the numbers. 7
b) What do you understand by pointer? Explain pointer arithmetic in brief. 8

6. a) Create a structure with the following data fields:

7

Emp_id	Name	Address	Salary	DOB		
				mm	dd	yy

Input records of 100 employee and display the record of those employee whose address is “Kathmandu”.

- b) Why do we need file pointer? Explain different file opening modes.

8

7. Write short notes on: (**Any two**)

2×5

- a) Storage Class
- b) Recursive function
- c) malloc(), calloc() and free()